

Total No. of Questions : 4]

SEAT No. :

PE140

[Total No. of Pages : 2

[6579]-473

T.E. (Computer Engg.) (Insem)

HONORS IN DATA SCIENCE

Data Science and Visualization

(2019 Pattern) (Semester-I) (310501)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2 and Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) What is Supervising Learning process. Elaborate with suitable example. [6]
b) Enlist different steps involved Data Science process and explain any one step in detail. [5]
c) What is Big Data and enlist its applications in detail? [4]

OR

- Q2)** a) What is exploratory Data Analysis in Data Science? Explain with suitable example. [6]
b) Differentiate between the Structured data and Unstructured data. [5]
c) Explain Unsupervised Machine Learning with its applications. [4]

- Q3)** a) Define the central tendency terms Mean, Median, Mode along with examples. [6]
b) Define Bayes Theorem with example. [5]
c) Define the statistical terms. [4]
i) Standardization
ii) Normalization

OR

P.T.O.

- Q4)** a) Calculate the mean, median, mode of the following data and comment on skewness of data. 17, 16, 25, 23, 22, 23, 28, 25, 25 [6]
- b) Define the Normal Distribution associated with probability. [5]
- c) Compare the variance and Standard Deviation with suitable examples. [4]

